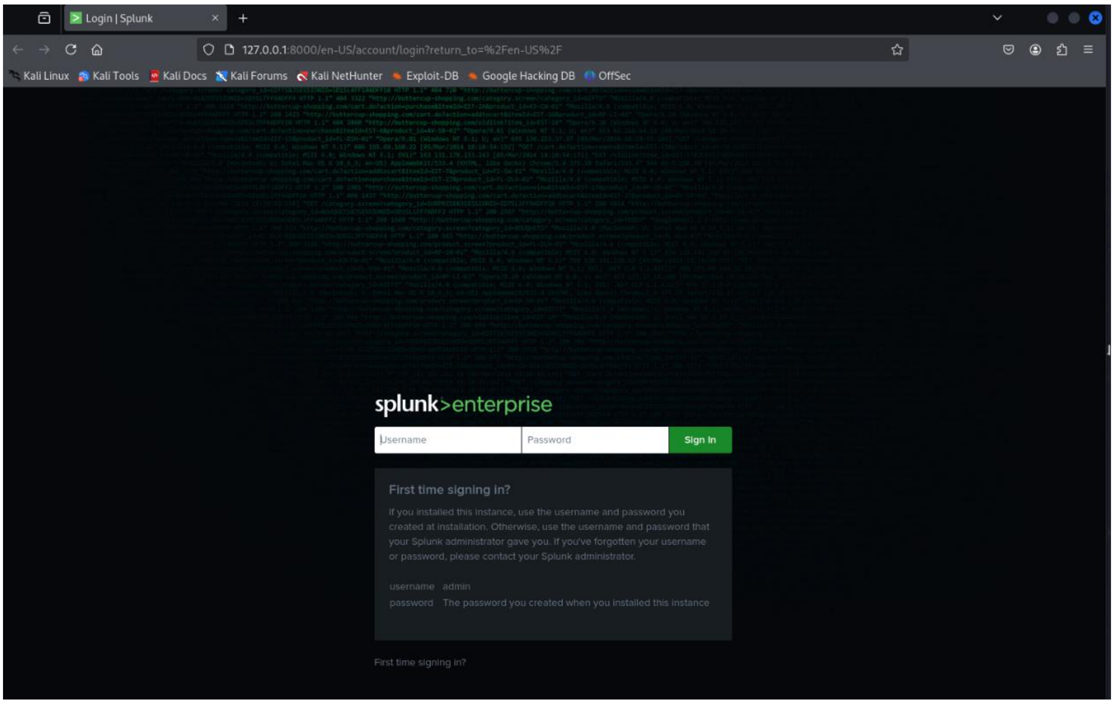
Phase 2: Visual Analysis with a SIEM Dashboard

Installed and configured Splunk on the Kali machine:

1. Download using this command
   wget -O splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb https://download.splunk.com/products/splunk/releases/9.3.2/linux/splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb
2. Then unpack using:
   sudo dpkg -i splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb
3. Start the splunk:
   sudo /opt/splunk/bin/splunk start --accept-license

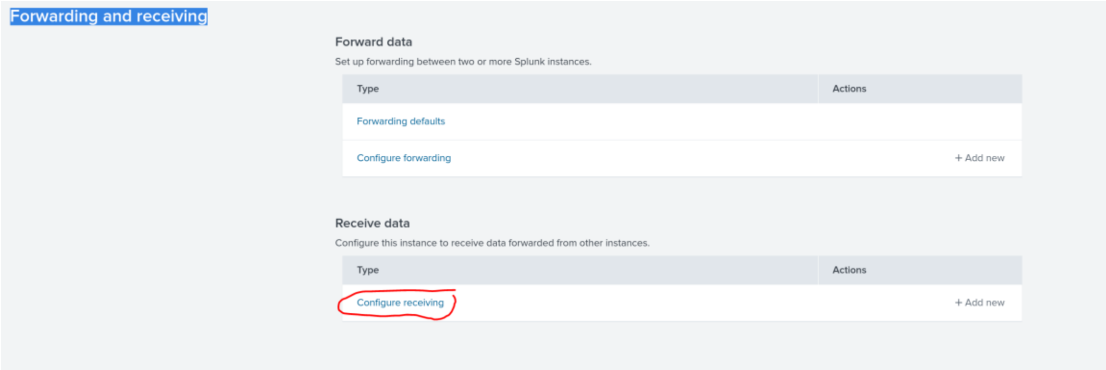
```
┌──(basim㉿Attacker)-[~/Desktop]
└─$ sudo /opt/splunk/bin/splunk start --accept-license
sudo: unable to resolve host Attacker: Name or service not known

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: asd
Password must contain at least:
   * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Generating RSA private key, 2048 bit long modulus
```

After registering we can open the web interface:

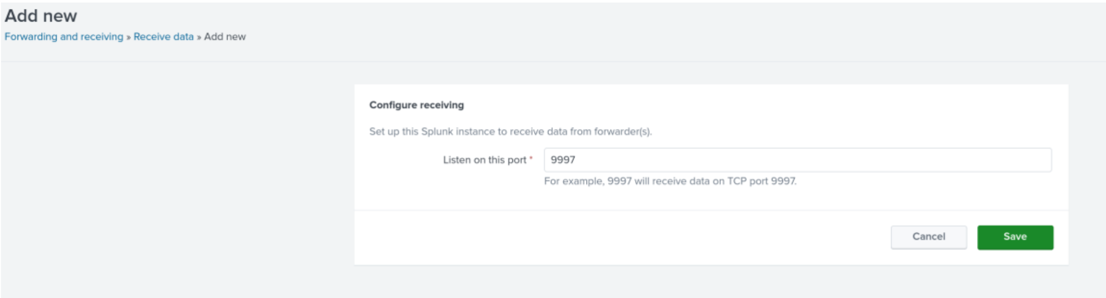


Make the splunk listen to port 9997:

1. Go to Settings → Forwarding and receiving
2. Click Configure receiving

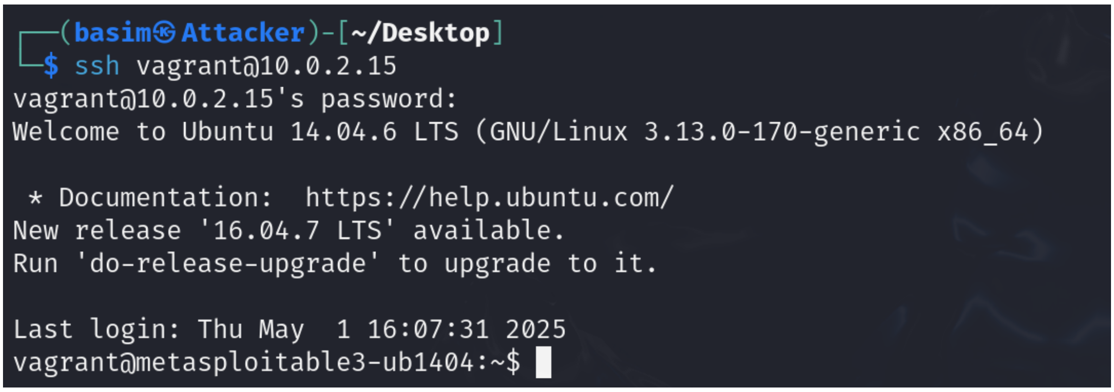


3. Click New Receiving Port and enter 9997



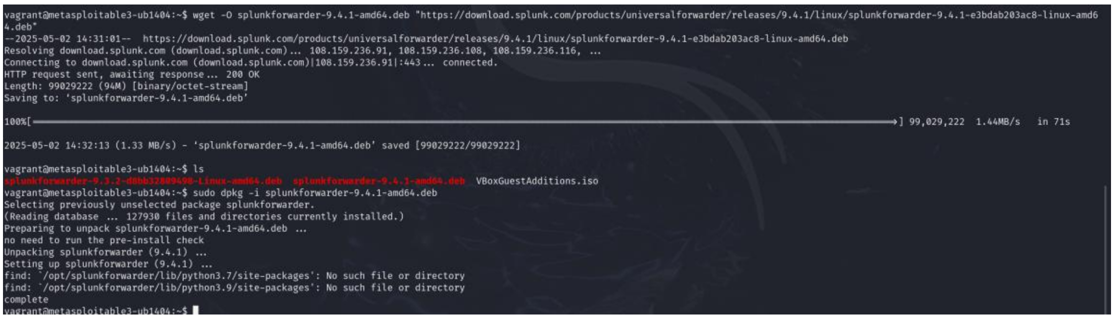
Configured the Splunk Universal Forwarder on Metasploitable3 to send logs to Splunk (port 9997).

ssh to the victim:



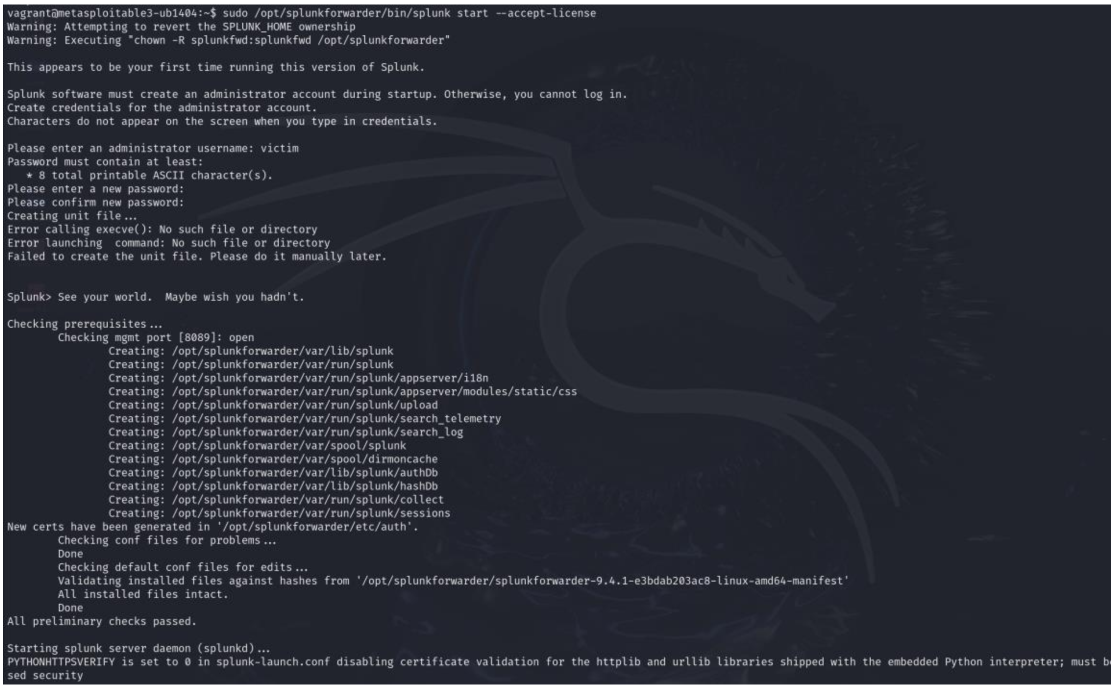
```
┌──(basim㉿Attacker)-[~/Desktop]
└─$ ssh vagrant@10.0.2.15
vagrant@10.0.2.15's password:
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 3.13.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu May  1 16:07:31 2025
vagrant@metasploitable3-ub1404:~$
```

Install **Splunk Universal Forwarder**:



```
vagrant@metasploitable3-ub1404:~$ wget -O splunkforwarder-9.4.1-amd64.deb "https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd6
4.deb"
--2025-05-02 14:31:01--  https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 108.159.236.91, 108.159.236.108, 108.159.236.116, ...
Connecting to download.splunk.com (download.splunk.com)|108.159.236.91|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 99029222 (94M) [binary/octet-stream]
Saving to: 'splunkforwarder-9.4.1-amd64.deb'

100%[==========================================================================>] 99,029,222  1.44MB/s   in 71s

2025-05-02 14:32:13 (1.33 MB/s) - 'splunkforwarder-9.4.1-amd64.deb' saved [99029222/99029222]

vagrant@metasploitable3-ub1404:~$ ls
splunkforwarder-9.3.2-d8bb32809498-Linux-amd64.deb  splunkforwarder-9.4.1-amd64.deb  VBoxGuestAdditions.iso
vagrant@metasploitable3-ub1404:~$ sudo dpkg -i splunkforwarder-9.4.1-amd64.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 127930 files and directories currently installed.)
Preparing to unpack splunkforwarder-9.4.1-amd64.deb ...
no need to run the pre-install check
Unpacking splunkforwarder (9.4.1) ...
Setting up splunkforwarder (9.4.1) ...
find: `/opt/splunkforwarder/lib/python3.7/site-packages': No such file or directory
find: `/opt/splunkforwarder/lib/python3.9/site-packages': No such file or directory
complete
vagrant@metasploitable3-ub1404:~$
```

Run it:



```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: victim
Password must contain at least:
   * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Error calling execve(): No such file or directory
Error launching  command: No such file or directory
Failed to create the unit file. Please do it manually later.


Splunk> See your world.  Maybe wish you hadn't.

Checking prerequisites...
        Checking mgmt port [8089]: open
                Creating: /opt/splunkforwarder/var/lib/splunk
                Creating: /opt/splunkforwarder/var/run/splunk
                Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
                Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
                Creating: /opt/splunkforwarder/var/run/splunk/upload
                Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
                Creating: /opt/splunkforwarder/var/run/splunk/search_log
                Creating: /opt/splunkforwarder/var/spool/splunk
                Creating: /opt/splunkforwarder/var/spool/dirmoncache
                Creating: /opt/splunkforwarder/var/lib/splunk/authDb
                Creating: /opt/splunkforwarder/var/lib/splunk/hashDb
                Creating: /opt/splunkforwarder/var/run/splunk/collect
                Creating: /opt/splunkforwarder/var/run/splunk/sessions
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and urllib libraries shipped with the embedded Python interpreter; must b
sed security
```

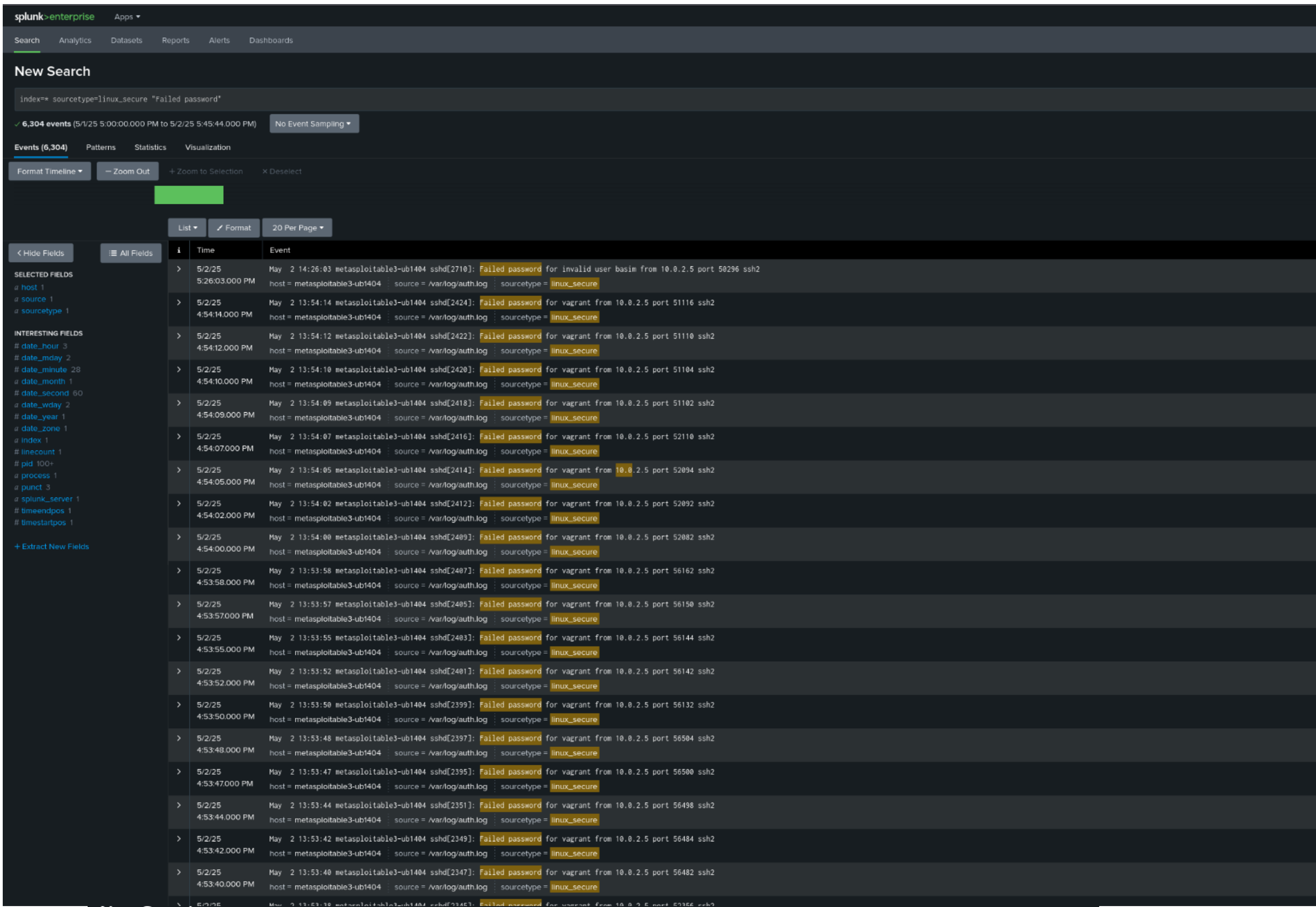
Monitored and uploaded logs like /var/log/auth.log.:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -index main -sourcetype linux_secure
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: victim
Password:
Added monitor of '/var/log/auth.log'.
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/syslog -index main -sourcetype linux_secure
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/syslog'.
vagrant@metasploitable3-ub1404:~$
```

Forward it to our server:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 10.0.2.5:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added forwarding to: 10.0.2.5:9997.
vagrant@metasploitable3-ub1404:~$
```

Run Splunk searches to detect failed and successful login attempts. (I will provide the csv files for the attempts in github)



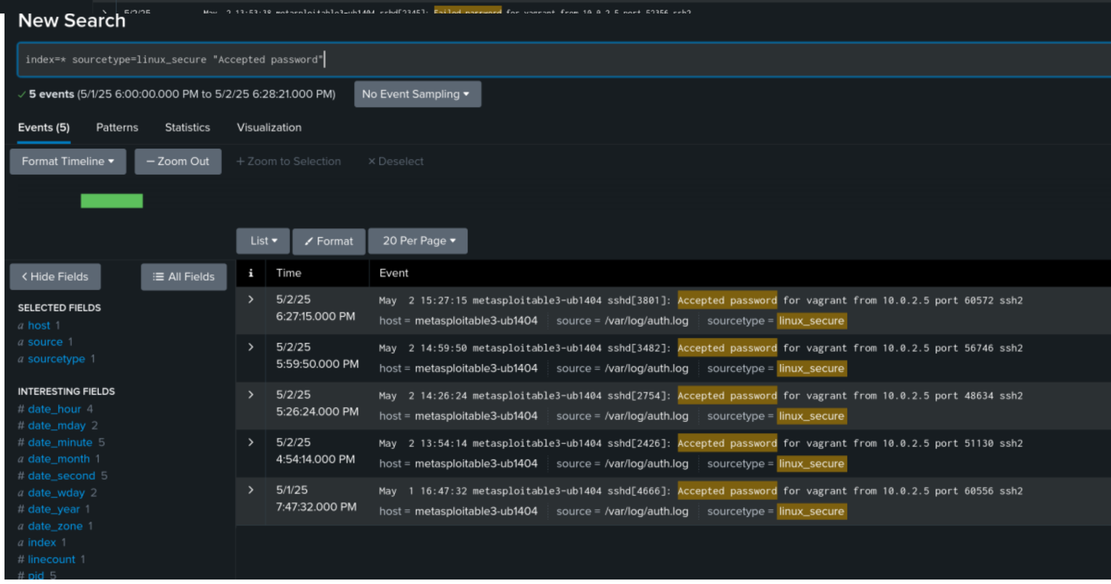
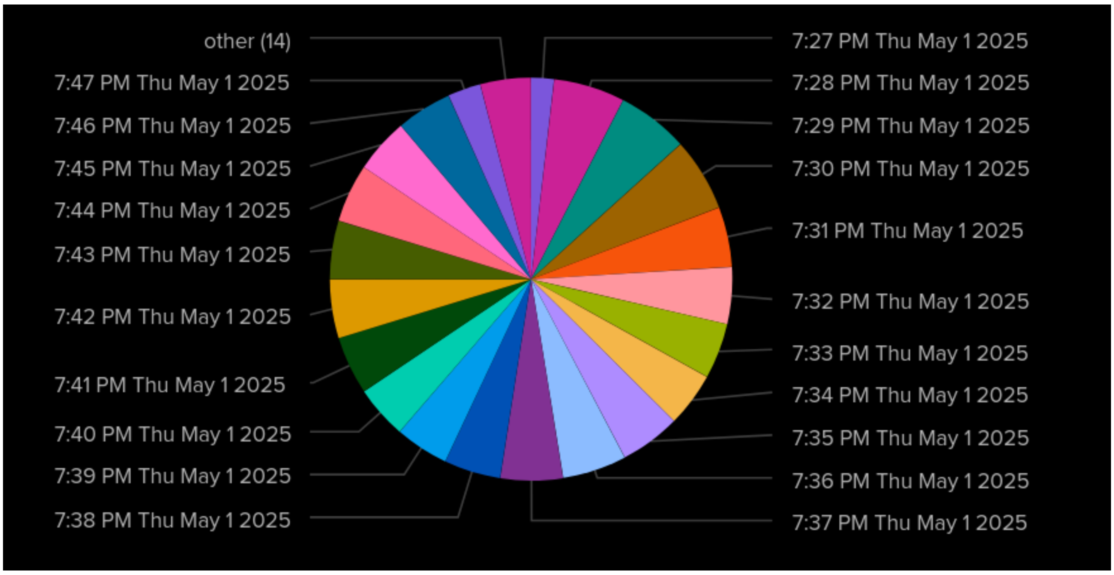


Chart of the attack pattern:



Conclusion:

Splunk successfully detected and visualized Hydra brute-force attempts on the victim machine. The data showed a high frequency of failed SSH logins from a single IP (the attacker).